ELTLS301 – Electrical lighting and power sockets installation

DATE: .../.../2025
PERIOD: .....

TECHNICAL SECONDARY SCHOOLS (TSS)

END OF TERM I EXAMINATIONS: 2025-2026

QUESTION PAPER

SECTOR: Energy

TRADE NAME: Electrical Technology

RQF LEVEL: 3

MODULE TITLE: **Electrical lighting and power sockets installation**

MARKS: **...../100**

DURATION: 3 HOURS

INSTRUCTIONS TO CANDIDATES:

This Exam paper is composed Sections (A, B, C). Follow the instructions given below, and answer the indicated questions in each section for a total of 100marks

Section **A**: 26 questions, all are **Compulsory**. **55 marks**
Section **B**: Five 5 questions, choose and answer any three (3). **30 marks**
Section **C**: Two 2 questions, choose and answer any one (1) **15 marks**

Allowed Materials: Blue pen, Ruler, Calculator, Pencil, (Fitting with the exam paper)

SECTION A: 26 questions, all are **Compulsory**. **55 marks**

MULTIPLE CHOICE (20 marks)

Choose the correct answer.

1. Electrical conduit is used for:
   a) Carrying water
   b) Protecting and routing electrical wiring
   c) Measuring voltage
   d) Supporting lighting fixtures
2. AC current is defined as:
   a) Current flowing only in one direction
   b) Current that periodically reverses direction
   c) Current that flows through batteries
   d) Current used only in communication systems
3. A junction box is used to:
   a) Control ON/OFF operations
   b) House electrical connections
   c) Secure cable clips
   d) Measure current
4. A wiring diagram shows:
   a) Physical layout of installation
   b) Electrical connection using symbols
   c) Location of fixtures in a building
   d) Specifications of materials
5. The size of a conductor is expressed in:
   a) Ohms
   b) mm^2 cross-sectional area
   c) Volts
   d) Watts
6. A PVC bendit is used to:
   a) Cut conduit
   b) Heat and soften PVC pipe for bending
   c) Test continuity
   d) Support trunking
7. Non-metallic (NM) sheathed wires are used for:
   a) Outdoor lighting
   b) In-house wiring
   c) High-voltage transmission
   d) Energy meter connections only
8. A ring terminal is best described as:
   a) A connector shaped like a fork
   b) A fully round terminal used with a screw
   c) A connector used for coax cables
   d) A switch terminal

9. Surface-mounted installation is:
   a) Installed inside the wall
   b) Installed outside the wall
   c) Installed underground
   d) Installed on a metallic roof
10. A pipe bender is used for:
   a) Stripping wires
   b) Twisting cables
   c) Bending conduit to shape
   d) Hammering fittings

TRUE OR FALSE (20 marks)

11. DC current flows only in one direction. _____
12. Cleaning means removing unwanted materials. _____
13. A switch box houses electrical connections. _____
14. Trunking is always circular in shape. _____
15. Built installation is performed inside walls. _____
16. A coaxial cable is used for high-frequency signal transmission. _____
17. A coupling is used to join two conduits. _____
18. Pipe curves are always 90°. _____
19. Wire strippers are used for removing insulation from conductors. _____
20. A socket outlet must be installed at least 450 mm above the floor. _____

FILL IN THE BLANKS (5 marks)

21. The _____ value is the maximum value of an alternating quantity.
22. A _____ is a tool used for measuring distances when installing conduits.
23. The document used in tendering that itemizes materials and costs is called a _____.
24. A _____ is a type of terminal shaped like a fork.
25. The process of changing conduit direction without accessories is called conduit _____.

MATCHING (10 marks)

Match column A with column B.

Column A

A. Electrical switch
B. Elbow 90°
C. Insulated conductor
D. Architectural diagram
E. Side cutter pliers

Column B

1. Used to cut and form loops on large conductors
2. Shows the location of each component in a building
3. Used to change conduit direction at a right angle
4. Interrupts or changes current flow
5. Wire covered with insulation

SECTION B: Five 5 questions, choose and answer any three (3). **30 marks**

1. Explain the parameters to consider when locating a power source in an installation site.
2. Describe the difference between built installation and surface-mounted installation.
3. List and explain three types of electrical diagrams.
4. Explain the procedure for chiselling during conduit placement.
5. Differentiate between PVC pipe and trunking.

SECTION C: Two 2 questions, choose and answer any one (1) **15 marks**

6. Describe the steps involved in measuring and cutting conduits.

7. Explain the difference between ring terminal, spade terminal, and bullet terminal.